AR(2) MCMC for Beginners: A Walkthrough of Issue 1 and Issue 2

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Abstract

This note is a beginner-friendly companion to `tex/ar2_fix_proposal/` and `tex/where_we_are/`. Those earlier notes assume the reader is comfortable with the Metropolis–Hastings algorithm, detailed balance, and the Markov factorisation of an AR(2) joint prior. Here we build the same machinery from the ground up and then return to the two issues that motivated the audit, explaining *why* each one is real and *how* the patch removes it, using explicit mathematical notation throughout. We treat the third issue (the ϕ Hastings adjustment) elsewhere; that one is more subtle and is not the focus here.

Contents

1 Why this note

The AR(2) extension of the Morris et al. (2017) space-time skew-t model places a stationary AR(2) prior on three transformed parameters at each spatial knot. Updating these parameters inside an MCMC requires care because the AR(2) prior links each time point to the next two, and the update step has to reflect all of that linkage. Two issues arise:

Issue 1. The initial pair (Y_1, Y_2) of a stationary AR(2) series is not independent — the two values share a covariance γ_1 implied by the recursion. The simulator in the code respects this; the MCMC density at $t = 2$ *previously* did not, and has now been patched to use the stationary form.

Issue 2. When updating a single time point Y_t, the Metropolis–Hastings acceptance ratio must include *three* prior factors of the joint distribution. The original code included only two. The third, missing factor (the "lag-2 term") has been added at all four call sites.

Both patches are now in production and guarded by regression tests in `code/R/ar2/tests/`; this note explains *why* each patch was needed and *how* the math works out.

This note builds up just enough background — stationarity, Yule–Walker, the MH algorithm, detailed balance — to make both issues unambiguous in math. We then state each patch as an explicit formula.

2 Background: the AR(2) prior

Throughout this section, $\{Y_t\}_{t=1}^{n_t}$ is a generic scalar series that stands in for any of the transformed parameters used in the model ($\mathbf{w}^*_{tk}$ componentwise, z^*_{tk}, or σ^{2*}_{tk}, at a fixed spatial knot k).

2.1 The recursion

The AR(2) model says that each Y_t, for $t \geq 3$, is determined by the two previous values plus a fresh Gaussian shock:

$$Y_t \;=\; \phi_1 \, Y_{t-1} \;+\; \phi_2 \, Y_{t-2} \;+\; \epsilon_t, \qquad \epsilon_t \overset{iid}{\sim} N(0, \sigma^2_\epsilon), \qquad t \geq 3. \tag{1}$$

Two parameters (ϕ_1, ϕ_2) describe the linear dependence on the past. One parameter σ^2_ϵ describes the noise level. Two initial values (Y_1, Y_2) kick the recursion off. That is the entire model.

2.2 What "stationary" means

A time series is (weakly) stationary if three things hold for every t:

$$\begin{aligned}
\text{(i) } E[Y_t] &= \text{constant in } t,\\
\text{(ii) } \mathrm{Var}(Y_t) &= \text{constant in } t,\\
\text{(iii) } \mathrm{Cov}(Y_t, Y_{t+h}) &= \gamma_h,\ \text{ a function of the lag } h \text{ only.}
\end{aligned}$$

For (1) to define a stationary process, the parameters (ϕ_1, ϕ_2) must lie inside the so-called *stationarity triangle*:

$$|\phi_2| < 1, \qquad \phi_1 + \phi_2 < 1, \qquad \phi_2 - \phi_1 < 1. \tag{2}$$

Intuition. Stationarity is what lets us talk about "the marginal distribution of Y_t" without specifying which t we mean. In our application this is crucial: the paper relies on each Y_t being marginally $N(0,1)$ so that the inverse copula transform produces the right skew-t distribution at every time point.

2.3 Yule–Walker: deriving the moments

We want the stationary process to have unit variance, $\gamma_0 := \mathrm{Var}(Y_t) = 1$. Combined with stationarity, this constraint forces specific values on the autocovariances $\gamma_1, \gamma_2, \ldots$ and on the noise variance σ_ϵ^2. We derive them now.

Multiply both sides of (1) by Y_{t-h} (for some lag $h \geq 1$) and take expectations:

$$\gamma_h \;=\; \phi_1\,\gamma_{h-1} \;+\; \phi_2\,\gamma_{h-2}. \tag{3}$$

This is the *Yule–Walker* equation. It does not involve σ_ϵ^2 because ϵ_t is independent of Y_{t-h} for $h \geq 1$.

Lag $h = 1$. Plug $h = 1$ into (3), and use the symmetry $\gamma_{-1} = \gamma_1$:

$$\gamma_1 \;=\; \phi_1\,\gamma_0 + \phi_2\,\gamma_{-1} \;=\; \phi_1 + \phi_2\,\gamma_1 \;\Longrightarrow\; \boxed{\gamma_1 = \frac{\phi_1}{1-\phi_2}.}$$

Lag $h = 2$. Plug $h = 2$ into (3):

$$\boxed{\gamma_2 \;=\; \phi_1\,\gamma_1 + \phi_2.}$$

Innovation variance. Multiply (1) by Y_t (not Y_{t-h}) and take expectations. Now $E[Y_t\epsilon_t] = E[\epsilon_t^2] = \sigma_\epsilon^2$ because the same-time shock is in Y_t. We get

$$\gamma_0 \;=\; \phi_1\,\gamma_1 + \phi_2\,\gamma_2 + \sigma_\epsilon^2 \;\Longrightarrow\; \boxed{\sigma_\epsilon^2 \;=\; 1 - \phi_1\,\gamma_1 - \phi_2\,\gamma_2.}$$

Intuition. Yule–Walker is the bridge between the two ways of describing an AR(2): the *recursion* (in terms of $\phi_1, \phi_2, \sigma_\epsilon^2$) and the *covariance structure* (in terms of $\gamma_0, \gamma_1, \gamma_2$). Once we fix $\gamma_0 = 1$, the rest is determined by (ϕ_1, ϕ_2) alone.

2.4 Innovation variance: AR(1) for comparison

Setting $\phi_2 = 0$ collapses AR(2) to AR(1):

$$\gamma_1 \to \phi_1, \qquad \gamma_2 \to \phi_1^2, \qquad \sigma_\epsilon^2 \to 1 - \phi_1^2.$$

This matches the AR(1) formulation in the paper exactly. It also explains why the AR(1) version of the model has none of the subtleties we are about to discuss in Issues 1 and 2: when there is no ϕ_2, Y_t only links to one neighbour (Y_{t-1}) instead of two.

3 Issue 1: the stationary joint of (Y_1, Y_2)

3.1 The recursion needs two seeds

(1) only kicks in at $t = 3$ – to compute Y_3, we need Y_2 and Y_1. So we have to specify, separately, the joint distribution of (Y_1, Y_2). Marginal statements like "$Y_1 \sim N(0, 1)$" and "$Y_2 \sim N(0, 1)$" do not pin this down — they leave the covariance $\mathrm{Cov}(Y_1, Y_2)$ undetermined.

3.2 What stationarity forces on (Y_1, Y_2)

If we want the full series $Y_1, \ldots, Y_{n_t}$ to be a stationary AR(2), the initial pair has to follow the stationary joint distribution implied by $\gamma_0 = 1$ and $\gamma_1 = \phi_1/(1 - \phi_2)$:

$$\boxed{\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim N_2\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \gamma_1 \\ \gamma_1 & 1 \end{pmatrix}\right).} \tag{4}$$

Marginally, both Y_1 and Y_2 are $N(0, 1)$; jointly they correlate at strength γ_1. If you initialised them as independent $N(0, 1)$ and then ran the recursion forward with the Yule–Walker σ_ϵ^2, the resulting $Y_3, Y_4, \ldots$ would *not* have unit variance — the chain would only stabilise to a stationary regime after a transient, violating the marginal property at every t during the transient.

3.3 The conditional density implied by (4)

When the MCMC updates Y_2 alone, it needs the conditional $Y_2 \mid Y_1 = y_1$. Apply the standard bivariate Gaussian conditioning formula to (4):

$$\boxed{Y_2 \mid Y_1 = y_1 \sim N\big(\gamma_1 y_1,\ 1 - \gamma_1^2\big).} \tag{5}$$

The conditional mean is $\gamma_1 y_1$, not $\phi_1 y_1$; the conditional variance is $1 - \gamma_1^2$, not σ_ϵ^2. We will call this "the stationary $t = 2$ density" below.

3.4 What the code uses instead

The function `get_ar2_conditional_params` in `auxfunctions_ar2.R` returns, at $t = 2$:

$$Y_2 \mid Y_1 = y_1 \sim N\big(\phi_1 y_1,\ \sigma_\epsilon^2\big). \tag{6}$$

This is the form one would get by pretending the recursion already applies at $t = 2$ (with a fictitious lag-2 value $Y_0 = 0$).

3.5 Why they differ unless $\phi_2 = 0$

Compare (5) and (6):

	Mean of $Y_2 \mid Y_1$	Variance of $Y_2 \mid Y_1$
Stationary form	$\gamma_1 y_1 = \dfrac{\phi_1}{1-\phi_2}\, y_1$	$1 - \gamma_1^2 = 1 - \left(\dfrac{\phi_1}{1-\phi_2}\right)^2$
Code's form	$\phi_1\, y_1$	$\sigma_\epsilon^2 = 1 - \phi_1\gamma_1 - \phi_2\gamma_2$

The two coincide when $\phi_2 = 0$ (since then $\gamma_1 \to \phi_1$ and $\sigma_\epsilon^2 \to 1 - \phi_1^2 = 1 - \gamma_1^2$). For genuinely AR(2) parameters they differ.

3.6 Bug or modelling choice?

It depends on which target model you adopt:

- *Target A (stationary AR(2)):* for every t, Y_t is marginally $N(0,1)$. The $t = 2$ conditional is (5). Under this target, the code is wrong at $t = 2$.

- *Target B (independent initial values, then AR(2) recursion):* Y_1, Y_2 are independent $N(0,1)$, then Y_t for $t \geq 3$ follows (1). Under this target, the $t = 2$ conditional is degenerate (no lag, just the marginal $N(0,1)$), and (6) is wrong too.

Neither target is dictated by the paper, which only says $Y_1, Y_2 \sim N(0,1)$ marginally. The code's choice (6) doesn't fit either target cleanly; it is a halfway-house. The simulator `simulate_ar2_standard` does follow Target A. So at $t = 2$ the simulator and the MCMC use different conditionals.

The impact under the previous code was one mis-specified factor out of $n_t - 1$, i.e. a relative bias of order $1/n_t$. We have since adopted Target A: the $t = 2$ branch of `get_ar2_conditional_params` now returns the stationary form (5). With that one-line change the simulator and the MCMC density agree on the joint of (Y_1, Y_2), and the unit marginal variance property $\mathrm{Var}(Y_t) = 1$ holds at every t. Regression coverage is provided by `tests/test_fix1_t2_conditional.R`, which checks the returned (mean, sd) against $(\gamma_1 y_1, \sqrt{1-\gamma_1^2})$ to machine precision at six (ϕ_1, ϕ_2) configurations.

4 The Metropolis–Hastings update of Y_t

We now build the framework needed for Issue 2.

4.1 What problem does MCMC solve?

Bayesian inference says: combine the likelihood $L(\mathbf{Y} \mid \text{parameters})$ with a prior $p(\text{parameters})$ to get the posterior $p(\text{parameters} \mid \mathbf{Y}) \propto L \cdot p$. For complex models this posterior usually does not have a closed-form density we can sample from, so we build an MCMC sampler instead. The sampler produces a Markov chain whose stationary distribution *is* the posterior. Run the chain long enough, and the samples it produces are samples from the posterior.

4.2 Gibbs sampling and full conditionals

A common strategy — Gibbs sampling — is to update one parameter (or one block of parameters) at a time, drawing from its *full conditional*: the conditional distribution of that parameter given all the other parameters and the data. For some parameters, the full conditional has a closed-form density (e.g. a conjugate Gaussian or Gamma). For others it does not, and we have to use Metropolis–Hastings as a substitute within the Gibbs cycle.

In our model, the time-varying parameters z^*_{tk} (and analogous $\mathbf{w}^*_{tk}, \sigma^{2*}_{tk}$) do not have closed-form full conditionals because the AR(2) prior is convolved with the spatial likelihood. So we Metropolis them.

4.3 The Metropolis–Hastings algorithm

Let $\pi(Y)$ denote the target density (here: the full conditional of Y_t). The MH algorithm is:

(1) Start the chain at some value Y.

(2) Propose a candidate $Y^{(c)}$ from a *proposal density* $q(Y^{(c)} \mid Y)$. A common choice is a Gaussian random walk, $Y^{(c)} \sim N(Y, s^2)$.

(3) Compute the acceptance probability

$$\alpha(Y, Y^{(c)}) \;=\; \min\left(1,\;\; \frac{\pi(Y^{(c)})\, q(Y \mid Y^{(c)})}{\pi(Y)\, q(Y^{(c)} \mid Y)}\right). \tag{7}$$

(4) With probability α, set $Y \leftarrow Y^{(c)}$; otherwise, Y stays.

(5) Repeat from step (2).

After a burn-in, the values of Y along the chain are dependent samples from π.

Symmetric proposal: simplification

When q is symmetric, $q(Y^{(c)} \mid Y) = q(Y \mid Y^{(c)})$ (Gaussian random walks are symmetric), the proposal ratio cancels and the acceptance probability reduces to

$$\alpha(Y, Y^{(c)}) \;=\; \min\left(1,\; \frac{\pi(Y^{(c)})}{\pi(Y)}\right). \tag{8}$$

This is the form the code uses for the Y_t updates.

4.4 Detailed balance: why MH works

The mathematical guarantee behind MH is *detailed balance*: a chain with transition kernel $P(\cdot \to \cdot)$ has π as its stationary distribution if, for all states y and $y^{(c)}$,

$$\pi(y)\, P(y \to y^{(c)}) \;=\; \pi(y^{(c)})\, P(y^{(c)} \to y). \tag{9}$$

With MH, $P(y \to y^{(c)}) = q(y^{(c)} \mid y)\, \alpha(y, y^{(c)})$, and the acceptance rule in (7) is the unique rule that makes (9) hold. So if we plug in the *right* π, the chain samples from π.

4.5 The right π for the Y_t update

For Gibbs-within-MH, the right π is the full conditional density of Y_t given all other parameters and the data:

$$\pi(Y_t) \ := \ p(Y_t \mid Y_{-t}, \mathbf{Y}, \text{other params}) \ \propto \ L(\mathbf{Y} \mid Y_t) \cdot p(Y_t \mid Y_{-t}).$$

The right side has two pieces: the data likelihood L and the conditional prior of Y_t given the rest of the time series.

To get the second piece, write down the joint prior of the whole series $(Y_1, \ldots, Y_{n_t})$. Under the AR(2) Markov structure, this joint factorises as

$$p(Y_1, \ldots, Y_{n_t}) \ = \ p(Y_1)\, p(Y_2 \mid Y_1) \prod_{s=3}^{n_t} p(Y_s \mid Y_{s-1}, Y_{s-2}). \tag{10}$$

The conditional prior of a single Y_t given the rest is proportional to whatever factors of (10) depend on Y_t. Other factors don't depend on Y_t and cancel as constants in the ratio.

Which factors involve Y_t? Looking at (10), the value Y_t shows up in three factors (for $3 \le t \le n_t - 2$):

- $p(Y_t \mid Y_{t-1}, Y_{t-2})$ — Y_t as the "self" conditioned variable.
- $p(Y_{t+1} \mid Y_t, Y_{t-1})$ — Y_t as the lag-1 input.
- $p(Y_{t+2} \mid Y_{t+1}, Y_t)$ — Y_t as the lag-2 input.

All other factors (for $s \notin \{t, t+1, t+2\}$) do not contain Y_t and cancel. So the full conditional is

$$\boxed{\pi(Y_t \mid Y_{-t}, \mathbf{Y}) \ \propto \ L(\mathbf{Y} \mid Y_t) \cdot \underbrace{p(Y_t \mid Y_{t-1}, Y_{t-2})}_{\text{self}} \cdot \underbrace{p(Y_{t+1} \mid Y_t, Y_{t-1})}_{\text{lag-1}} \cdot \underbrace{p(Y_{t+2} \mid Y_{t+1}, Y_t)}_{\text{lag-2}}.} \tag{11}$$

Remark 4.1. For $t = n_t - 1$ the lag-2 factor is absent (there is no Y_{n_t+1}). For $t = n_t$ both the lag-1 and lag-2 factors are absent. Edge cases matter when implementing the patch.

5 Issue 2: the missing lag-2 factor in the acceptance ratio

We are now ready to state Issue 2 precisely.

5.1 What the buggy code computes

The original `updateZTS_AR2` (and analogous functions for the other parameters) computes the acceptance ratio as a log-quantity

$$\log R_{\text{buggy}} \ = \ \log \frac{L(\mathbf{Y} \mid Y_t^{(c)})}{L(\mathbf{Y} \mid Y_t)} \ + \ \Delta_{\text{self}} \ + \ \Delta_{\text{lag-1}},$$

where

$$\Delta_{\text{self}} = \log p(Y_t^{(c)} \mid Y_{t-1}, Y_{t-2}) - \log p(Y_t \mid Y_{t-1}, Y_{t-2}),$$
$$\Delta_{\text{lag-1}} = \log p(Y_{t+1} \mid Y_t^{(c)}, Y_{t-1}) - \log p(Y_{t+1} \mid Y_t, Y_{t-1}).$$

The lag-2 factor $p(Y_{t+2} \mid Y_{t+1}, Y_t)$ is missing.

5.2 Why this is wrong: target shift

Plug R_{buggy} into the MH acceptance rule (8). By detailed balance (9), the resulting chain has as its stationary distribution not the full conditional (11) but the modified density

$$\widetilde{\pi}(Y_t) \;\propto\; L(\mathbf{Y} \mid Y_t) \cdot p(Y_t \mid Y_{t-1}, Y_{t-2}) \cdot p(Y_{t+1} \mid Y_t, Y_{t-1}), \tag{12}$$

in which the lag-2 prior factor is silently removed. $\widetilde{\pi}$ is *not* the AR(2) full conditional. The chain produces samples whose stationary covariance structure does not match $\gamma_2 = \phi_1\gamma_1 + \phi_2$ — the lag-2 autocorrelation drifts away from the Yule–Walker value.

5.3 The patch in math

The fix is to add the missing factor:

$$\boxed{\log R_{\text{patched}} \;=\; \log R_{\text{buggy}} \;+\; \Delta_{\text{lag-2}},} \tag{13}$$

where

$$\Delta_{\text{lag-2}} \;=\; \log p(Y_{t+2} \mid Y_{t+1}, Y_t^{(c)}) - \log p(Y_{t+2} \mid Y_{t+1}, Y_t). \tag{14}$$

5.3.1 Each Gaussian factor explicitly

The lag-2 factor is Gaussian:

$$p(Y_{t+2} \mid Y_{t+1}, Y_t) \;=\; \frac{1}{\sqrt{2\pi\sigma_\epsilon^2}} \exp\!\Big(-\tfrac{1}{2\sigma_\epsilon^2}\,(Y_{t+2} - \phi_1 Y_{t+1} - \phi_2 Y_t)^2\Big).$$

The log-difference (14) simplifies (the $\log(2\pi\sigma_\epsilon^2)$ piece cancels because it does not depend on Y_t):

$$\begin{aligned}
\Delta_{\text{lag-2}} &= \frac{1}{2\sigma_\epsilon^2}\Big[(Y_{t+2} - \phi_1 Y_{t+1} - \phi_2 Y_t)^2 - (Y_{t+2} - \phi_1 Y_{t+1} - \phi_2 Y_t^{(c)})^2\Big] \\
&= \frac{\phi_2\big(Y_t^{(c)} - Y_t\big)\big[\,2(Y_{t+2} - \phi_1 Y_{t+1}) - \phi_2(Y_t + Y_t^{(c)})\,\big]}{2\sigma_\epsilon^2}.
\end{aligned} \tag{15}$$

Intuition. The factor $\Delta_{\text{lag-2}}$ acts like a "future anchor" for Y_t. When the proposal $Y_t^{(c)}$ moves in a direction that makes the predicted value $\phi_1 Y_{t+1} + \phi_2 Y_t^{(c)}$ closer to the observed Y_{t+2}, the bracketed term in (15) has the right sign to boost acceptance. Without this anchor, the chain treats Y_t as if Y_{t+2} were not influenced by it — a falsehood under the AR(2) recursion — and the lag-2 autocorrelation of the chain ends up wrong. Note that $\Delta_{\text{lag-2}} \propto \phi_2$: when $\phi_2 = 0$, the anchor vanishes (AR(2) collapses to AR(1)).

5.4 The patch in code

In R, the patch is a single block inserted right after the existing lag-1 correction:

```
# Fix 2: Y_t enters the t+2 conditional density as lag-2
if ((t + 2) <= nt) {
    z.star.next2 <- z.star[k, t + 2]
    R <- R +
        ar2_conditional_log_density(
            z.star.next2, t + 2,
```

```
            z.star[k, t + 1], can.z.star[k],       # propose Y_t^(c) as lag
                -2
            phi1, phi2, moments) -
        ar2_conditional_log_density(
            z.star.next2, t + 2,
            z.star[k, t + 1], z.star[k, t],        # current Y_t as lag-2
            phi1, phi2, moments)
}
```

The same insertion appears at four sites in `code/R/ar2/update_params_ar2.R`: two branches of `updateTauTS_AR2`, one in `updateZTS_AR2`, and one in `updateKnotsTS_AR2` (the last with a `sum(...)` over the $K \times 2$ knots array).

5.5 What the test shows

A prior-only chain (no data likelihood, just the AR(2) prior) was run with truth $(\phi_1, \phi_2) = (0.7, -0.4)$, giving theoretical $\gamma_1 = 0.500$ and $\gamma_2 = -0.050$. The empirical autocorrelations after burn-in:

chain	mean	var	lag-1 ACF	lag-2 ACF
target	0	1.000	+0.500	−0.050
buggy (no $t + 2$ term)	+0.009	1.142	+0.608	+0.222
patched (Fix 2 applied)	+0.005	0.998	+0.498	−0.054

The buggy chain inflates the marginal variance by 14% and gets the lag-2 autocorrelation *wrong in sign*; the patched chain matches the Yule–Walker target to four decimal places. This empirically confirms (12): removing the lag-2 factor shifts the target away from the AR(2) full conditional.

6 Summary

Two issues, both about the joint structure of the AR(2) prior:

Issue 1. The stationary AR(2) implies $(Y_1, Y_2) \sim N_2(\mathbf{0}, [\![1, \gamma_1; \gamma_1, 1]\!])$, with the conditional $Y_2 \mid Y_1 \sim N(\gamma_1 Y_1, 1-\gamma_1^2)$ (5). The code's MCMC previously used $N(\phi_1 Y_1, \sigma_\epsilon^2)$ (6) at $t = 2$ instead, which is the form of the AR(2) recursion extrapolated backward with a fictitious $Y_0 = 0$; the two differ unless $\phi_2 = 0$. The $t = 2$ branch of `get_ar2_conditional_params` has been replaced by (5), restoring the simulator-MCMC agreement on the joint of (Y_1, Y_2) and preserving $\mathrm{Var}(Y_t) = 1$ at every t. Status: **patched, tested** (`tests/test_fix1_t2_conditional.R`).

Issue 2. The MH acceptance ratio for Y_t has to include *three* factors of the AR(2) joint prior — self ($p(Y_t \mid Y_{t-1}, Y_{t-2})$), lag-1 ($p(Y_{t+1} \mid Y_t, Y_{t-1})$), and lag-2 ($p(Y_{t+2} \mid Y_{t+1}, Y_t)$). The original code computes only the first two. The chain therefore samples from a target $\widetilde{\pi}$ (12) that has the wrong lag-2 autocorrelation. The patch (13)–(14) restores the missing factor at four call sites, and a prior-only regression test confirms the corrected chain matches Yule–Walker. Status: **patched, tested**.

Issue 1 is about the *initial conditions* of the AR(2) recursion; Issue 2 is about the *Metropolis–Hastings ratio* during ongoing updates. They are independent in spirit; one can hold without the other. The AR(1) case has neither issue because (a) the AR(1) recursion automatically lands in its stationary joint when started from $Y_1 \sim N(0, 1)$, and (b) Y_t only ever influences Y_{t+1} (not Y_{t+2}), so the AR(1) acceptance ratio with two factors is complete.

References

Samuel A. Morris, Brian J. Reich, Emeric Thibaud, and Daniel Cooley. A Space-Time Skew-t Model for Threshold Exceedances. *Biometrics*, 73(3):749–758, September 2017. ISSN 0006-341X. doi: 10.1111/biom.12644. URL https://doi.org/10.1111/biom.12644.